

PROJECT PRESENTATION

Cloud-Based Road Damage Reporting and Monitoring System

Turning citizen reports into faster, data-driven road repairs



Presented by [Name] · [Institution / Course] · [Date]

Why we need it

THE PROBLEM

Manual reporting

Phone, letter or walk-in complaints with no exact location or photo.

Scattered records

Each department keeps its own data, so duplicates go unnoticed.

No feedback loop

Citizens rarely learn if a pothole was scheduled or repaired.

Result: damaged roads stay open longer, raising accident risk.

FOUR GOALS FOR THE PLATFORM

1

Easy reporting

Log damage with a photo and GPS in a few taps.

2

One cloud platform

Store every report in a single scalable system.

3

Smart prioritisation

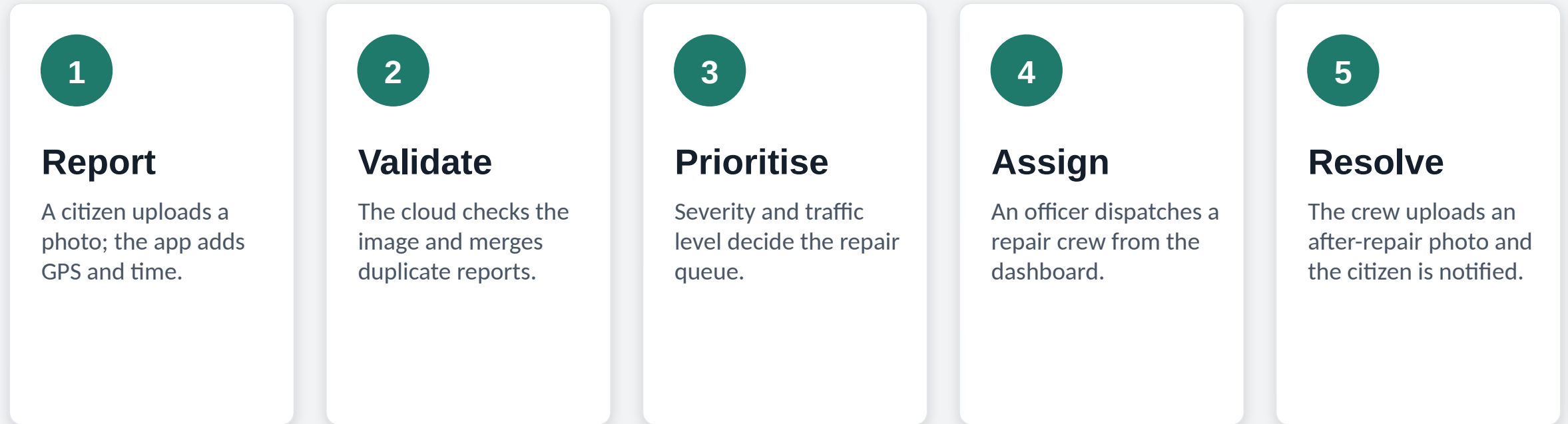
Rank repairs by severity, traffic and repeat reports.

4

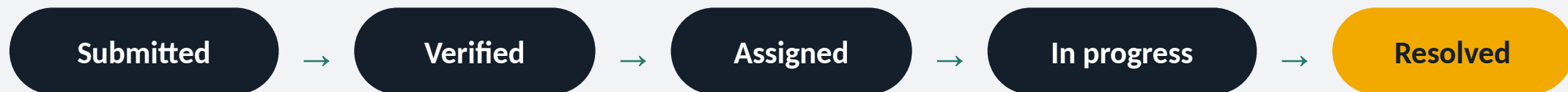
Transparent tracking

Show live status from submission to completed repair.

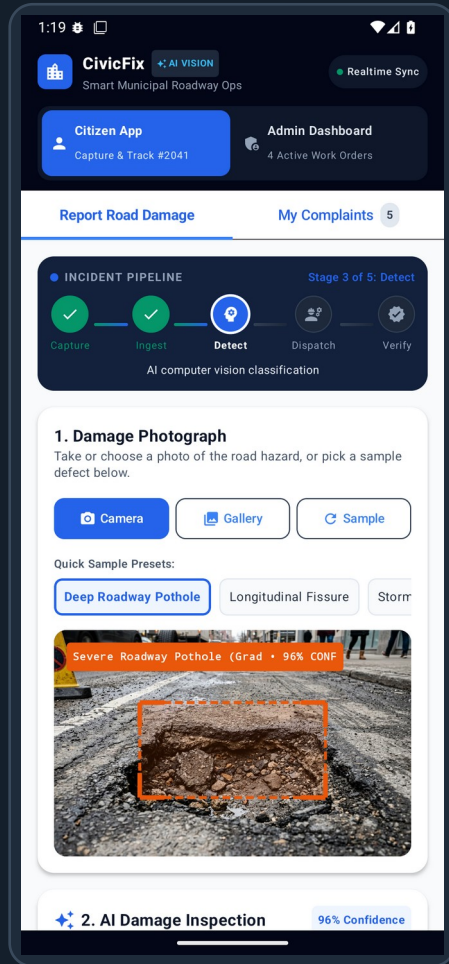
How the system works



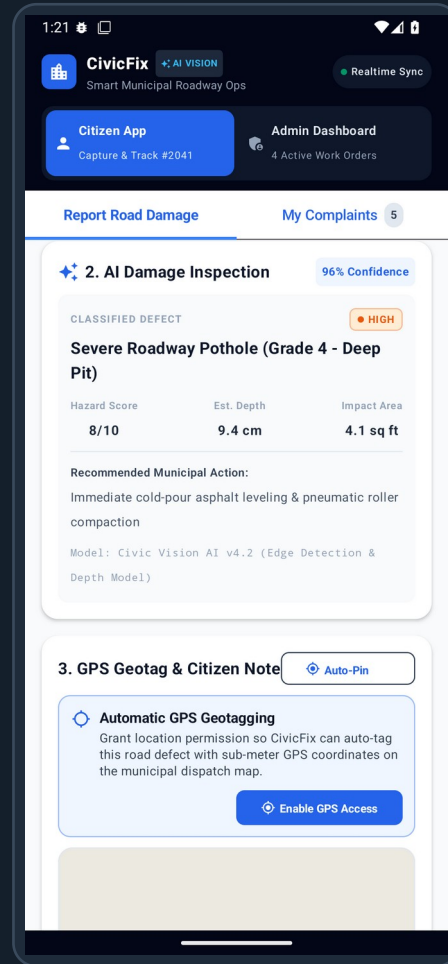
Report status lifecycle



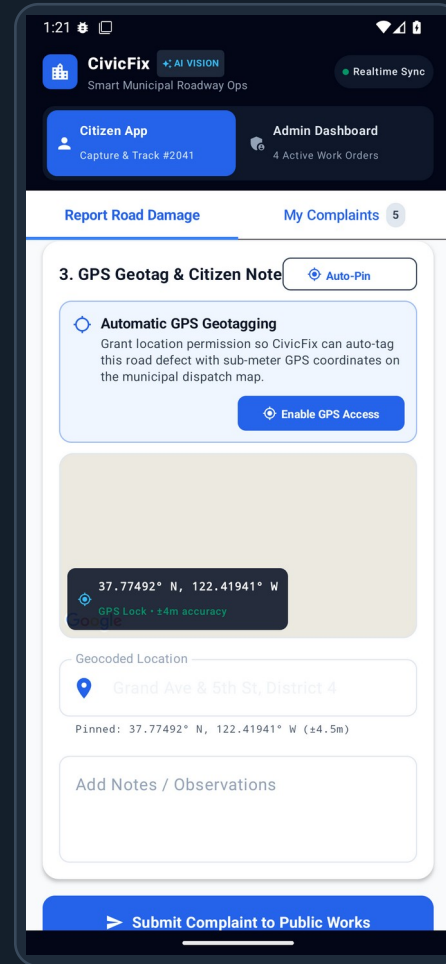
How the app works



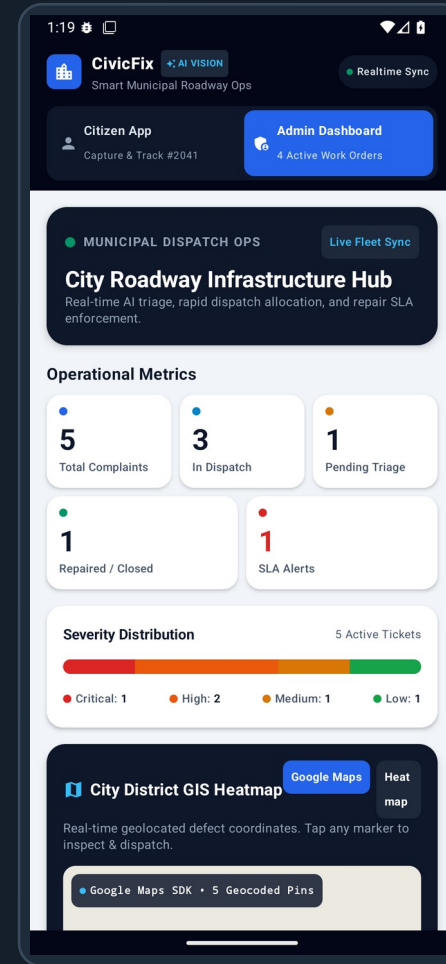
1. Capture
AI detects the damage



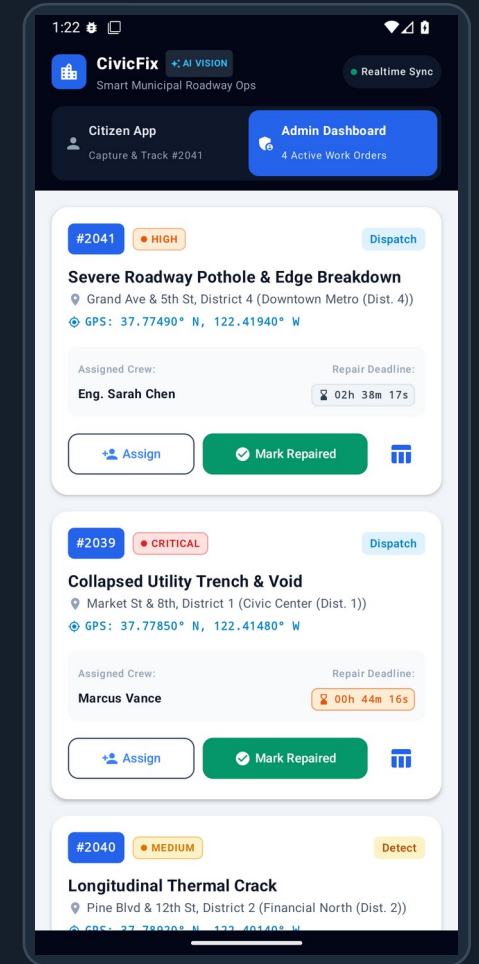
2. Inspect
Severity, depth, action



3. Geotag
GPS pin and notes



4. Triage
Admin metrics and map



5. Dispatch
Assign crews, track SLA

Four cloud layers, one tech stack

LAYER 1

Client

Citizen app, web portal, authority dashboard, field-crew app

LAYER 2

Access

API gateway, authentication, load balancer, push alerts

LAYER 3

Processing

Report service, AI detection, duplicate and priority engine

LAYER 4

Storage

Image storage, geospatial database, cache, logs and backups

PROPOSED TECH STACK

Mobile app

Flutter or React Native

Web dashboard

React + map library

Backend API

FastAPI or Node.js

AI model

CNN / YOLO detection

Database

PostgreSQL + PostGIS

Cloud services

AWS S3, Lambda, Cognito

What the platform offers

1 Photo and GPS reports

Location and time captured automatically.

2 AI damage detection

Spots potholes and cracks and rates severity.

3 Duplicate merging

Nearby reports of one defect become one case.

4 Priority scoring

Severity, road type and traffic set the priority.

5 Live status alerts

Citizens are notified at every repair stage.

6 Analytics dashboard

Maps and charts reveal hotspots and repair times.

One dashboard for every road

Live severity map · Filters by area, type, status and date · Crew assignment and progress tracking · Trend charts

Benefits, risks and mitigations

Citizens

Report in a few taps, follow status live, get notified when fixed.

Authorities

One clear view of damage, data-based priorities, fewer duplicate inspections.

Society

Safer roads, lower vehicle repair costs, more trust through open tracking.

Challenge	Mitigation
Fake or blurry reports	AI validation, GPS and time checks
Weak roadside internet	Offline capture, upload later
Privacy of location data	Encryption and role-based access
Cost as reports grow	Serverless auto-scaling, storage rules
Low citizen participation	Simple app and status alerts

Where the system can grow

1 Detection from vehicle dashcams and sensors

2 Drone and satellite imagery for road surveys

3 Predictive maintenance before roads fail

4 Smart-city and municipal integration

5 Multilingual and voice-based reporting

Thank You

One cloud platform connecting citizens, crews and planners to fix roads faster.

Questions and discussion

[Name] • [Email]